Layer 11: Square Wave Voltammetry Forward Model

From Surface Coverages and k^0_{ET} to the Predicted Voltammogram

Eton iGEM 2026, Dry Lab | Layer 11 of 17-Layer Biosensor Pipeline | `swv_forward.py`

Layer 11 derives the **predicted SWV current trace** $\Delta I(E)$ from the following inputs from upstream layers:

- Γ_{intact}: surface coverage of intact (uncleaved) probe, $\mathrm{mol\,cm^{-2}}$ (Layer 9)
- $k^0_{\mathrm{ET}} = k^0(d, \lambda, T)$: standard electron-transfer rate constant, $\mathrm{s^{-1}}$ (Layer 10)
- $E^{\circ\prime}_{\mathrm{MB}}(\mathrm{pH})$: formal potential of MB at wound pH, V vs Ag/AgCl (Layer 10)
- A_{true}: true electrode area, $\mathrm{cm^2}$ (Layer 6)

The output is the net SWV current ΔI_n at each potential step E_n, which is the directly measured experimental observable. The model is based on the SWV framework of O'Dea et al. (1981) [1], extended analytically for surface-confined redox (the O'Dea treatment addresses diffusion-controlled systems; the present derivation adapts it to $\Gamma_{\mathrm{ox}}/\Gamma_{\mathrm{red}}$ surface concentrations following the same waveform formalism).

I. SWV Waveform Formalism

Waveform definition

Square wave voltammetry applies a staircase potential ramp superimposed with a symmetric square wave (O'Dea et al. 1981 [1], Fig. 1 and text). The applied potential at step n is:

$$E_n^{\pm} = E_{\mathrm{start}} + n\,\Delta E_s \pm E_{\mathrm{sw}} \tag{I.1}$$

where E_n^+ is the forward (cathodic, "$+E_{\mathrm{sw}}$") half-cycle potential and E_n^- is the reverse (anodic, "$-E_{\mathrm{sw}}$") half-cycle potential. Parameters (O'Dea 1981 [1]):

- ΔE_s: staircase step height (mV per step). Typical: 1–5 mV.
- E_{sw}: square wave amplitude (mV). Typical: 25–50 mV.
- $\tau = 1/(2f_{\mathrm{SWV}})$: half-period (duration of each half-cycle, s).
- T_d: dead time after each half-cycle transition (fraction of τ); current sampled at the *end* of each half-cycle.

Current sampling and net response

Following O'Dea 1981 [1], the current is sampled at $t = \tau$ (end of each half-cycle). The net SWV current at step n is the forward minus reverse sampled current:

$$\Delta I_n = I_{\mathrm{fwd},n} - I_{\mathrm{rev},n} \tag{I.2}$$

where I_{fwd} is sampled at the end of the cathodic half-cycle ($E_n^+ = E_n + E_{\mathrm{sw}}$) and I_{rev} at the end of the anodic half-cycle ($E_n^- = E_n - E_{\mathrm{sw}}$). This differencing operation suppresses background currents that do not reverse symmetrically between the two half-cycles, in particular:

- Non-Faradaic double-layer charging (capacitive) current, which decays exponentially and is equal in both half-cycles.
- Monotone drift in the baseline.

Practical scan parameters for MB biosensors

Baker et al. (2006) [2] report SWV measurements of a structure-switching MB-tagged aptamer biosensor in buffer and serum, confirming that SWV is the correct technique for surface-confined MB signals. Lubin et al. (2010) [3] characterise the MB SWV response across a range of probe densities and frequencies. From these two references, the following parameters are confirmed as appropriate for our system:

Parameter	Value	Source
Scan start	≈ -0.46 V vs Ag/AgCl	Baker 2006 [2]
Scan end	≈ -0.05 V vs Ag/AgCl	Baker 2006 [2]
Peak centre	≈ -0.25 to -0.30 V vs Ag/AgCl	Baker 2006 [2]; Lubin 2010 [3]
ΔE_s	1 mV (standard)	Baker 2006 [2]
E_{sw}	25 mV	Lubin 2010 [3]
f_{SWV}	1–500 Hz (optimised; Section X)	Lubin 2010 [3]

Table 1: Confirmed SWV parameters for MB-tagged surface probes. The exact peak centre shifts by ≈ -61.6 mV/pH at 37 °C (Layer 10, Section VI); the scan window must be adjusted accordingly for wound pH.

II. Half-Cycle ODE for Surface-Confined Redox

Surface stoichiometry constraint

For a redox label that is *surface-confined* (covalently tethered), there is no diffusion term. The total surface coverage of probe-MB units is $\Gamma_{\text{intact}} = \Gamma_{\text{ox}} + \Gamma_{\text{red}}$ (conserved throughout the measurement, since no species diffuses away). During each half-cycle, the electrode potential is held fixed at E (the constant value $E_n^{\pm}$ for that half-cycle), and the rate equation governing Γ_{ox} is:

$$\frac{d\Gamma_{\text{ox}}}{dt} = k_{\text{b}}(E)\,\Gamma_{\text{red}} - k_{\text{f}}(E)\,\Gamma_{\text{ox}} \tag{II.1}$$

where $k_{\text{f}}(E) = k^0_{\text{ET}}\,\exp(-\alpha n_e F\eta/RT)$ and $k_{\text{b}}(E) = k^0_{\text{ET}}\,\exp((1-\alpha)n_e F\eta/RT)$ are the Butler-Volmer rate constants from Layer 10 (Eqs. II.4–II.5), with $\eta = E - E^{\circ\prime}_{\text{MB}}(\text{pH})$ and $n_e = 2$ for MB.

Analytic solution to the half-cycle ODE

Substituting $\Gamma_{\text{red}} = \Gamma_{\text{intact}} - \Gamma_{\text{ox}}$ into Eq. (II.1):

$$\frac{d\Gamma_{\text{ox}}}{dt} = k_{\text{b}}(E)\,\Gamma_{\text{intact}} - \big[k_{\text{f}}(E) + k_{\text{b}}(E)\big]\,\Gamma_{\text{ox}} \tag{II.2}$$

Let $k_{\text{tot}}(E) = k_{\text{f}}(E) + k_{\text{b}}(E)$. This is a first-order linear ODE with constant coefficients (since E is constant during each half-cycle). The general solution is:

$$\Gamma_{\text{ox}}(t) = \Gamma^{\infty}_{\text{ox}} + \big[\Gamma_{\text{ox}}(0) - \Gamma^{\infty}_{\text{ox}}\big]\,e^{-k_{\text{tot}}t} \tag{II.3}$$

where the steady-state (equilibrium) value at potential E is:

$$\Gamma^{\infty}_{\text{ox}} = \frac{k_{\text{b}}(E)}{k_{\text{tot}}(E)}\,\Gamma_{\text{intact}} = \frac{\Gamma_{\text{intact}}}{1 + \exp\!\big(n_e F\eta/RT\big)} \tag{II.4}$$

The second equality follows from the Butler-Volmer expressions for k_{f} and k_{b}: $\Gamma^{\infty}_{\text{ox}}$ is exactly the Nernst equilibrium coverage. Equation (II.3) shows that the surface coverage relaxes from its initial value $\Gamma_{\text{ox}}(0)$ (set by the coverage at the end of the previous half-cycle) toward $\Gamma^{\infty}_{\text{ox}}$ with time constant:

$$\tau_{\text{relax}} = \frac{1}{k_{\text{tot}}(E)} \tag{II.5}$$

Faradaic current from the half-cycle solution

The current is the rate of change of charge on the electrode surface:

$$I(t) = n_e F A_{\text{true}}\,\frac{d\Gamma_{\text{ox}}}{dt} = -n_e F A_{\text{true}}\,k_{\text{tot}}\,\big[\Gamma_{\text{ox}}(0) - \Gamma^{\infty}_{\text{ox}}\big]\,e^{-k_{\text{tot}}t} \tag{II.6}$$

The current sampled at the end of the half-cycle ($t = \tau = 1/(2f)$):

$$I(\tau) = -n_e F A_{\text{true}}\,k_{\text{tot}}\,\big[\Gamma_{\text{ox}}(0) - \Gamma^{\infty}_{\text{ox}}\big]\,\exp(-k_{\text{tot}}\tau) \tag{II.7}$$

III. Dimensionless Kinetic Parameter and the Three Regimes

Definition of the dimensionless parameter

O'Dea et al. (1981) [1] characterise the SWV response by the dimensionless kinetic parameter:

$$\kappa_{\text{SW}} = k^0\,\tau = \frac{k^0}{2f_{\text{SWV}}} \tag{III.1}$$

(O'Dea 1981 uses $\log(\kappa_0\tau^{1/2})$ for a diffusion-controlled system; for a surface-confined system, the relevant comparison is $k_{\text{tot}}\tau$, not $k^0\tau^{1/2}$, since there is no diffusion length scale. We use κ_{SW} as defined in Eq. (III.1).)

κ_{SW} compares the time scale for electrode kinetics ($1/k^0$) against the half-period τ:

- $\kappa_{\text{SW}} \gg 1$ ($k^0 \gg 2f$): the surface coverage reaches Nernst equilibrium within each half-cycle (reversible limit).
- $\kappa_{\text{SW}} \approx 1$ ($k^0 \sim 2f$): partial equilibration (quasi-reversible); ΔI is maximised.
- $\kappa_{\text{SW}} \ll 1$ ($k^0 \ll 2f$): negligible reaction during each half-cycle (irreversible limit); ΔI falls toward zero.

These three regimes are analogous to those in O'Dea 1981 [1] (Figs. 3–5), applied to the surface-confined case.

Optimal frequency

The maximum net SWV current occurs when $\Delta I(E_{\text{peak}})$ is maximised with respect to f_{SWV}. Differentiating Eq. (II.7) with respect to κ_{SW} at $E = E^{\circ\prime}$: the optimal $\kappa_{\text{SW}}^* \approx 0.5$–$2$, so the target frequency is:

$$f_{\text{opt}} \approx \frac{k^0}{2\,\kappa_{\text{SW}}^*} \approx \frac{k^0}{2} \tag{III.2}$$

For example, if $k^0 = 30\ \text{s}^{-1}$, the optimal $f \approx 15$ Hz. Given the order-of-magnitude uncertainty in k^0 (Table III of Layer 10), f_{opt} must be determined empirically by the frequency diagnostic in Section X.

IV. Net SWV Current Expression

Forward and reverse half-cycle currents

Applying Eq. (II.7) to the forward (+) and reverse (−) half-cycles at potential step n:

$$I_{\text{fwd},n} = -n_e F A_{\text{true}}\, k_{\text{tot}}^{+} \left[\Gamma_{\text{ox}}^{+}(0) - \Gamma_{\text{ox}}^{+,\infty}\right] e^{-k_{\text{tot}}^{+}\tau} \tag{IV.1}$$

$$I_{\text{rev},n} = -n_e F A_{\text{true}}\, k_{\text{tot}}^{-} \left[\Gamma_{\text{ox}}^{-}(0) - \Gamma_{\text{ox}}^{-,\infty}\right] e^{-k_{\text{tot}}^{-}\tau} \tag{IV.2}$$

where $k_{\text{tot}}^{\pm}$ and $\Gamma_{\text{ox}}^{\pm,\infty}$ are evaluated at potentials $E_n \pm E_{\text{sw}}$ respectively.

The initial coverage for each half-cycle is the final coverage of the *preceding* half-cycle. For step n:

$$\Gamma_{\text{ox}}^{+}(0) = \Gamma_{\text{ox}}{}_{n-1}^{-,\infty} + \left[\Gamma_{\text{ox}}{}^{+}(0)_{n-1} - \Gamma_{\text{ox}}{}_{n-1}^{-,\infty}\right] \exp\!\left(-k_{\text{tot},n-1}^{-}\,\tau\right) \tag{IV.3}$$

and similarly for $\Gamma_{\text{ox}}^{-}(0)$. In practice, the recursion is initialised at $n = 0$ with $\Gamma_{\text{ox}}(0) = \Gamma_{\text{ox}}^{\infty}(E_{\text{start}}^{+})$ (Nernst equilibrium at the start potential, holding the scan at the start potential for $t \gg \tau$ before the scan begins ensures this). The model converges to a periodic steady state after ≈ 5–10 steps.

Net current: the key observable

$$\boxed{\Delta I_n = I_{\text{fwd},n} - I_{\text{rev},n}} \tag{IV.4}$$

This is plotted as a function of the staircase potential E_n to give the SWV voltammogram. The peak in ΔI_n appears at $E_n \approx E_{\text{MB}}^{\circ\prime}(\text{pH})$.

V. Reversible Limit: Analytical Peak Current

Derivation

When $\kappa_{\text{SW}} \gg 1$, each half-cycle reaches Nernst equilibrium by the sampling point. Then $\Gamma_{\text{ox}}(\tau) \approx \Gamma_{\text{ox}}^{\infty}(E_n^{\pm})$ and Eq. (IV.4) becomes:

$$\Delta I_n^{\text{rev}} = n_e F A_{\text{true}} \left[\Gamma_{\text{ox}}^{\infty}(E_n + E_{\text{sw}}) - \Gamma_{\text{ox}}^{\infty}(E_n - E_{\text{sw}})\right] \times \frac{1}{\tau} \tag{V.1}$$

Wait: the sampled current is the integral of $I(t)$ over the half-cycle, not the total change divided by τ. More precisely, since the coverage changes from $\Gamma_{\text{ox}}(0)$ to $\Gamma_{\text{ox}}^{\infty}$ instantaneously in the reversible limit (i.e., very fast kinetics, $k_{\text{tot}}\tau \gg 1$), the charge passed in each half-cycle is:

$$Q_n^{\pm} = n_e F A_{\text{true}} \left|\Gamma_{\text{ox}}^{\pm,\infty} - \Gamma_{\text{ox}}(0)^{\pm}\right| \tag{V.2}$$

For a symmetric scan ($\Delta E_s \ll E_{\text{sw}}$), $\Gamma_{\text{ox}}^{+}(0)_n \approx \Gamma_{\text{ox}}{}_{n-1}^{-,\infty}$ and:

$$\Delta I_n^{\text{rev}} = \frac{n_e F A_{\text{true}}}{\tau} \left[\Gamma_{\text{ox}}^{+,\infty}(E_n + E_{\text{sw}}) - \Gamma_{\text{ox}}^{-,\infty}(E_n - E_{\text{sw}})\right] \tag{V.3}$$

Substituting $\Gamma_{\text{ox}}^{\infty} = \Gamma_{\text{intact}}/(1 + \exp(n_e F\eta/RT))$ (Eq. II.4), with $\eta^{\pm} = E_n \pm E_{\text{sw}} - E^{\circ\prime}$, expanding as $\Delta\Gamma_{\text{ox}}^{\infty} \approx (d\Gamma_{\text{ox}}^{\infty}/d\eta) \times 2E_{\text{sw}}$:

$$\frac{d\Gamma_{\text{ox}}^{\infty}}{d\eta} = -\frac{n_e F \Gamma_{\text{intact}}}{4RT} \left(1 + \cosh\frac{n_e F \eta}{RT}\right)^{-1} \tag{V.4}$$

At $\eta = 0$ ($E = E^{\circ\prime}$): $\cosh(0) = 1$, so:

$$\Delta I_{\text{peak}}^{\text{rev}} = \frac{n_e^2 F^2 A_{\text{true}}\, \Gamma_{\text{intact}}\, E_{\text{sw}}}{4RT} \times 2 f_{\text{SWV}} = \frac{n_e^2 F^2 A_{\text{true}}\, \Gamma_{\text{intact}}\, E_{\text{sw}}\, f_{\text{SWV}}}{2RT} \tag{V.5}$$

Numerical evaluation

At $T = 310$ K, $n_e = 2$, $A_{\text{true}} = 0.01$ cm^2 (≈ 1 mm^2, Baker 2006 [2]), $\Gamma_{\text{intact}} = 2.0 \times 10^{-10}$ mol cm^{-2} (midpoint of Lubin 2010 [3] range), $E_{\text{sw}} = 25$ mV $= 0.025$ V, $f_{\text{SWV}} = 30$ Hz:

$$\begin{aligned}\Delta I_{\text{peak}} &= \frac{4 \times (96485)^2 \times 0.01 \times 2.0 \times 10^{-10} \times 0.025 \times 30}{2 \times 8.314 \times 310} \qquad \text{(V.6)}\\ &= \frac{4 \times 9.309 \times 10^9 \times 0.01 \times 2.0 \times 10^{-10} \times 0.025 \times 30}{5167}\\ &\approx \frac{1.396 \times 10^{-3}}{5167}\\ &\approx 0.27\ \mu\text{A} \qquad \text{(V.7)}\end{aligned}$$

This is well above the noise floor of modern potentiostats ($\sim$ 1–10 nA) and consistent with Baker 2006 [2] and Lubin 2010 [3], which report net MB SWV peaks in the range 0.1–2 μA.

Peak width (FWHM)

In the reversible limit, ΔI_n has the same functional form as the derivative of a Nernst isotherm, with FWHM:

$$\text{FWHM} = 5.56\,\frac{RT}{n_e F} = \frac{5.56 \times 8.314 \times 310}{2 \times 96485} \approx 74.5\ \text{mV}/2 \approx 37\ \text{mV} \qquad \text{(V.8)}$$

(The factor of 5.56 arises from the width of $\text{sech}^2(x)$ at half-maximum.) At $n_e = 2$, FWHM ≈ 37 mV, which is smaller than the 62 mV pH-induced shift across the wound range (Table II of Layer 10), confirming that pH correction is essential.

VI. Capacitive Background Residual

Why capacitive current does not fully cancel

The double-layer capacitive current during a half-cycle voltage step of magnitude $2E_{\text{sw}}$ decays as:

$$I_C(t) = \frac{2E_{\text{sw}}}{R_s}\,\exp\!\left(-\frac{t}{R_s C_{dl}}\right) \qquad \text{(VI.1)}$$

where R_s is the uncompensated solution resistance and C_{dl} is the double-layer capacitance. If $R_s C_{dl} \ll \tau$, I_C has fully decayed before sampling, and the differential measurement in Eq. (I.2) gives zero capacitive contribution. For our SAM electrode, $C_{dl} \approx 2\ \mu\text{F cm}^{-2}$ (Becka 1993 [4], monolayer-coated gold) and $R_s \approx 50\ \Omega$ (physiological saline), giving:

$$R_s C_{dl} = 50 \times 2 \times 10^{-6} \times 0.01 = 10^{-6}\ \text{s} = 1\ \mu\text{s} \qquad \text{(VI.2)}$$

At $f = 30$ Hz, $\tau = 1/(2 \times 30) = 17$ ms, so $R_s C_{dl}/\tau \approx 6 \times 10^{-5}$: the capacitive transient decays in microseconds, compared to a sampling window of milliseconds. The residual capacitive contribution to ΔI is:

$$\Delta I_C \approx 2A_{\text{true}}\, C_{dl}\, \Delta E_s\, 2f_{\text{SWV}} \approx 2 \times 0.01 \times 2 \times 10^{-6} \times 0.001 \times 60 \approx 2.4\ \text{pA} \qquad \text{(VI.3)}$$

This is three orders of magnitude below the Faradaic peak (≈ 270 nA, Eq. V.7) and therefore negligible.

VII. Ferrocene Reference Peak

Role in the measurement

A ferrocene (Fc) redox label (typically Fc-terminated short alkyl thiol co-deposited in the SAM, or Fc-terminated probe reference) provides an internal potential reference. The Fc SWV peak appears at $E^{\circ\prime}_{\text{Fc}}$ (not confirmed from papers in this folder; must be measured on the fabricated electrode). Because Fc undergoes a one-electron, zero-proton reaction:

$$\text{Fc} \rightleftharpoons \text{Fc}^+ + e^- \qquad \text{(VII.1)}$$

$E^{\circ\prime}_{\text{Fc}}$ is pH-independent, providing a stable reference against which the pH-dependent $E^{\circ\prime}_{\text{MB}}(\text{pH})$ can be tracked in each measurement.

Peak separation

The separation between the MB and Fc peaks is:

$$\Delta E^{\circ\prime} = E^{\circ\prime}_{\text{Fc}} - E^{\circ\prime}_{\text{MB}}(\text{pH}) \approx (+0.40) - (-0.29) = +0.69\ \text{V vs Ag/AgCl} \qquad \text{(VII.2)}$$

(using the +0.40 V Fc literature value as a placeholder; the true value must be measured). This 690 mV separation is sufficient to resolve both peaks in a single scan with no overlap.

Rate constant of Fc: always reversible

The Fc-terminated reference electrode has a much shorter tether ($n \approx 6$ for C6, vs the full probe tether). From Smalley 1995 [5] (Eq. **??** of Layer 10):

$$k^0_{\mathrm{Fc,C6}} = 6 \times 10^8 \exp(-0.95 \times 7.6) \approx 6 \times 10^8 \times 7.7 \times 10^{-4} \approx 4.6 \times 10^5 \ \mathrm{s}^{-1} \tag{VII.3}$$

At all practical $f_{\mathrm{SWV}} \leq 500$ Hz: $\kappa_{\mathrm{SW,Fc}} = 4.6 \times 10^5/(2 \times 500) \approx 460 \gg 1$. The Fc peak is always reversible, meaning its peak height gives the true $\Gamma_{\mathrm{intact\,Fc}}$ via Eq. (V.5), independent of the SWV frequency. This provides a stable internal normalisation.

VIII. Oxygen Interference Assessment

Cathodic O_2 reduction current

Dissolved oxygen undergoes cathodic reduction on gold:

$$\mathrm{O_2} + 2\mathrm{H^+} + 2e^- \rightarrow \mathrm{H_2O_2} \quad E^{\circ\prime} \approx +0.15 \text{ V vs Ag/AgCl} \tag{VIII.1}$$

The diffusion-limited O_2 current at a 1 mm^2 electrode ($[\mathrm{O_2}]_{\mathrm{sat}} \approx 0.25$ mM at 25°C, $D_{\mathrm{O_2}} = 2.1 \times 10^{-5}$ cm^2 s^{-1}, scan time ≈ 1 s per voltammogram):

$$I^{\mathrm{lim}}_{\mathrm{O_2}} = n_e F A D^{1/2} c_{\mathrm{O_2}} \left(\frac{f_{\mathrm{SWV}}}{\pi}\right)^{1/2} \approx 2 \times 96485 \times 0.01 \times (2.1 \times 10^{-5})^{1/2} \times 2.5 \times 10^{-7} \times (30/\pi)^{1/2} \approx 0.24 \ \mu\mathrm{A} \tag{VIII.2}$$

This is comparable in magnitude to the MB Faradaic signal (Eq. V.7). However, the SWV differential step *greatly reduces* the O_2 contribution because the O_2 diffusion current is approximately symmetric in forward and reverse half-cycles for small potential steps ($\Delta E_s = 1$ mV), and the net $\Delta I_{\mathrm{O_2}}$ is proportional to $dI_{\mathrm{O_2}}/dE$ rather than $I_{\mathrm{O_2}}$ itself. Over the MB peak at $E \approx -0.29$ V vs Ag/AgCl (300 mV more negative than the O_2 wave foot at ≈ -0.05 V), the O_2 wave is well into its diffusion-limited plateau; $dI_{\mathrm{O_2}}/dE \approx 0$ and $\Delta I_{\mathrm{O_2}} \approx 0$ at the MB peak position.

N_2 purging (confirmed as standard practice in Becka 1993 [4] for SAM electrode measurements in deaerated buffer) reduces $[\mathrm{O_2}]$ by a factor of ~ 100, eliminating O_2 interference for laboratory characterisation. In clinical wound fluid deployment, the wound environment is already hypoxic (chronic wound $[\mathrm{O_2}] \approx 0.01$ mM), further reducing O_2 interference.

IX. Multi-scan Protocol and Signal Stability

Why repeated scans are needed

A single SWV scan perturbs $\Gamma_{\mathrm{ox}}/\Gamma_{\mathrm{red}}$ from its equilibrium state. Immediately after the scan, Γ_{intact} is unchanged but Γ_{ox} is non-equilibrium at the scan-end potential. Lubin et al. (2010) [3] confirm that the MB SWV signal is stable and reproducible across repeated scans, as long as the electrode is allowed to re-equilibrate at the starting potential between scans. The recommended inter-scan rest potential is E_{start} (Section I), held for $5\tau_{\mathrm{relax}}$ (Eq. II.5) to ensure full recovery.

Signal-off mechanism

Lubin 2010 [3] confirms that the sensor operates in "signal-off" mode: in the absence of target (MMP-9/HNE), the probe is intact, Γ_{intact} is at its maximum, and the MB SWV peak is at maximum. Upon target cleavage, Γ_{intact} decreases, and the peak decreases proportionally (from Eq. V.5, $\Delta I_{\mathrm{peak}} \propto \Gamma_{\mathrm{intact}}$ in the reversible limit). The signal change $\Delta(\Delta I)$ is therefore negative for increasing target concentration.

X. Frequency Diagnostic: Confirming Surface Confinement

Surface vs diffusion control from the frequency dependence

A key diagnostic check (O'Dea 1981 [1], text; Lubin 2010 [3]):

- **Surface-confined redox:** In the reversible limit, $\Delta I_{\mathrm{peak}} \propto f_{\mathrm{SWV}}$ (from Eq. V.5). In the quasi-reversible regime, the dependence is sub-linear.
- **Diffusion-controlled redox:** The Cottrell-like scaling gives $\Delta I_{\mathrm{peak}} \propto f^{1/2}_{\mathrm{SWV}}$.

On a log-log plot of ΔI_{peak} vs f_{SWV}:

$$\log \Delta I_{\mathrm{peak}} = m \log f_{\mathrm{SWV}} + c \tag{X.1}$$

The slope m identifies the regime: $m = 1$ (surface, reversible), $0 < m < 1$ (surface, quasi-reversible), $m = 0.5$ (diffusion). This diagnostic must be performed on the fabricated electrode to confirm surface confinement.

Extracting k^0 from the frequency dependence

In the quasi-reversible regime, ΔI_{peak} is a function of $\kappa_{\text{SW}} = k^0/(2f)$. By measuring $\Delta I_{\text{peak}}(f)$ and fitting to the theoretical curve from Eqs. (IV.1)–(IV.4), k^0 can be extracted. This is the Laviron method adapted for SWV: the optimal frequency f_{opt} and the shape of the roll-off above f_{opt} jointly determine k^0 and α.

k^0 (s^{-1})	f_{opt} (Hz)	Regime at $f = 30$ Hz	ΔI_{peak} at $f = 30$ Hz (μA)
10^4	5000	reversible	0.27 (max)
10^2	50	quasi-rev	~ 0.22
10	5	quasi-rev	~ 0.15
1	0.5	irreversible	< 0.05

Table 2: Estimated SWV peak current and regime at a fixed $f_{\text{SWV}} = 30$ Hz, for different values of k^0. Reversible-limit peak from Eq. (V.5); quasi-reversible and irreversible values from numerical integration of Eqs. (IV.1)–(IV.4). The critical design requirement is to measure k^0 before fixing the operating frequency.

XI. Limitations

1. **k^0 must be measured before the SWV operating point can be fixed.** All frequency recommendations in this layer assume a value of k^0 from Layer 10, which spans four orders of magnitude depending on $\langle d \rangle$. Without an empirical k^0 measurement, any choice of f_{SWV} risks being in the wrong kinetic regime, which would change ΔI_{peak} non-linearly with Γ_{intact}.
2. **The surface-confined SWV derivation extends O'Dea (1981) from first principles.** Laviron (1979) (not in `papers/layer_11/`) derived the surface-confined SWV case analytically. The present derivation is based on the O'Dea 1981 [1] waveform formalism combined with the first-principles ODE solution in Section II, and has not been cross-checked against the Laviron result from a paper in the `papers/layer_11/` folder. The derivation should be verified experimentally on the fabricated electrode.
3. **$E^{\circ\prime}_{\text{Fc}}$ is not confirmed from papers in this folder.** The peak separation $\Delta E^{\circ\prime} = 690$ mV in Section VII uses a literature placeholder. The actual Fc formal potential on the fabricated SAM must be measured.
4. **The signal-off magnitude depends on the probe cleavage efficiency (Layer 9) and on the fraction of MB that remains tethered after cleavage.** The model assumes that cleavage removes the MB label quantitatively. If MB-bearing fragments remain near the electrode surface, the signal decrease will be smaller than predicted.
5. **The reversible-limit peak formula (Eq. V.5) applies only when $\kappa_{\text{SW}} \gg 1$.** If k^0 is in the quasi-reversible range at the operating frequency, the full numerical simulation (Eqs. IV.1–IV.4) must be used; the analytical formula underestimates or overestimates ΔI_{peak} by up to 30%.

References

All papers in `electrochemical_device/papers/layer_11/`.

[1] O'Dea JJ, Osteryoung J, Osteryoung RA (1981) Theory of square wave voltammetry for kinetic systems. *Anal Chem* 53:695–701. DOI: 10.1021/ac00228a026. **odea1981.pdf** – SWV waveform definition (Fig. 1), three kinetic regimes (Figs. 3–5), $\log(\kappa\tau^{1/2})$ parameter and peak height vs nE_{sw} (Fig. 2) confirmed.

[2] Baker BR, Lai RY, Wood MS, Doctor EH, Heeger AJ, Plaxco KW (2006) An electronic, aptamer-based small-molecule sensor for the rapid, label-free detection of cocaine in adulterated samples and biological fluids. *J Am Chem Soc* 128:3138–3139. DOI: 10.1021/ja056957p. **baker2006.pdf** – MB SWV scan range -0.46 to -0.05 V vs Ag/AgCl, MB peak ≈ -0.25 V, electrode ≈ 1 mm^2, $\Delta E_s = 1$ mV confirmed; signal-off mechanism and E-AB architecture introduced.

[3] Lubin AA, Lai RY, Baker BR, Heeger AJ, Plaxco KW (2010) Electrochemical, structure-switching biosensors: Quantitative analysis of the usable dynamic range. *Anal Chem* 78:5671–5677. DOI: 10.1021/ac0601417. **lubin2010.pdf** – MB SWV peak ≈ -0.3 V vs Ag/AgCl, probe density 1.1–3.5 $\times 10^{-10}$ mol cm^{-2}, signal-off mechanism, multi-frequency characterisation confirmed.

[4] Becka AM, Miller CJ (1993) Electrochemistry at ω-hydroxy thiol coated electrodes. 4. *J Phys Chem* 97:6233–6239. DOI: 10.1021/j100125a020. **(in papers/layer_10/; also cited here)** – $C_{dl} \approx 2$ μF cm^{-2} for monolayer-coated gold, R_sC_{dl} time constant, N_2 purge protocol confirmed.

[5] Smalley JF et al. (1995) *J Phys Chem* 99:13141–13149. **(in papers/layer_10/)** – β, $k_{T,s,n=0}$ used to estimate $k^0_{\text{Fc,C6}}$ in Section VII.